

Atomic Theories and Timeline

Solid Sphere Model

Plum Pudding Model

Nuclear Model

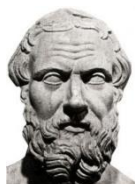
Planetary Model

Quantum Model

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640 BC

Democritus



He proposed that all forms of matter can be divided into tiny indivisible particles and called these particles as "atomos", which means "indivisible".

1803

John Dalton



He created the very first atomic theory. He viewed atoms as tiny solid balls.

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1904

J.J. Thompson



He showed that atom was made of even smaller things and discovered the electrons.

1911

Ernest Rutherford



He showed that atoms have a positively charged nucleus in the center and are mostly empty space. He discovered the proton and the nucleus.

1913

Niels Bohr



He improved Rutherford's model and proposed that electrons move around the nucleus in specific layers or shells.

1926

Erwin Schrodinger



He stated that electrons do not move in set paths around the nucleus but in waves. It is impossible to know the exact location of the electrons; instead, a "cloud of probability" called orbitals, in which we are more likely to find an electron.

1932

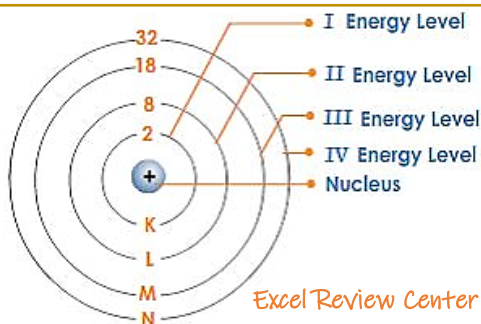
James Chadwick



He discovered the particles with no charge, called neutrons.

The Bohr Model

A further improvement of the Rutherford model, this atomic model still proposes that the atom consists of a positively charged core where most of the mass is concentrated. He added that electrons can only travel around this core or nucleus in certain orbits that corresponds to the energy of the electron. These orbits are called energy levels or orbitals. Electrons can only move from one orbital to another by losing or by gaining energy. The orbitals at which the electrons are found are designated by letters. The orbital "k" is the energy level that is closest to the nucleus. These orbits will not hold the same number of electrons. For instance, the first shell can hold 2 electrons only, the second shell can hold up to 8 electrons, the third shell up to 18, and so on. As a general rule, each shell can hold $2n^2$ electrons where n is the number of the orbit.



Quantum Numbers

One important part of the study of chemistry is understanding how electrons behave in an atom and how electrons interact among themselves between atoms and molecules. The properties of an atom's electron configuration depend on the 4 quantum numbers. Think of quantum numbers as addresses of the electrons:

1. The principal quantum number, n

This is also called the energy level quantum number. The lower the energy, the closer the electron is to the source of attraction. The higher its energy, the further out it goes. Hence, electrons found on lower energy levels are closer to the nucleus than those that have higher energy levels. It also follows that the higher the energy level, the higher is the value of n , the radius of the electron orbital also increases. The value of n can only take integer values (i.e. from 1, 2, 3...). Moreover, larger value of n denotes a bigger orbit and higher

energy. All electrons in an atom with the same value of n are said to belong to the same principle energy level.

2. The azimuthal quantum number, l

This is also called the angular quantum number or the orbital angular momentum quantum number. Each value of n (first quantum number) has multiple values of l ranging from 0 to $n-1$. This determines the shape of the electron cloud or the orbital within a shell. There are names for each values of l listed below:

l	sublevel
0	s
1	p
2	d
3	f

Quantum Numbers continuation...

3. The magnetic quantum number, m

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The third quantum number describes the orientation of orbitals within a sublevel. It does not affect the orbital energy. Only integer value between $-l$ and $+l$ are allowed. The number of m values within a sublevel is the number of orbital within a sublevel.

l	sublevel	possible values of m	number of orbitals in this sublevel
0	s	0	1
1	p	-1, 0, +1	3
2	d	-2, -1, 0, +1, +2	5
3	f	-3, -2, -1, 0, +1, +2, +3	7

4. The spin quantum number, s

This is the fourth quantum number. There are only two possible values for s : $+1/2$ and $-1/2$. These are also referred to as "spin up" or "spin down". This quantum number is used to explain behavior of individual electrons as if they were spinning clockwise or counterclockwise.

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